MC 501 Describing Data in R rubric

100 points. Due Monday, November 23, 11:59 PM.

How this is scored. Add the points from each row; the total is out of 100. A first attempt is graded forgivingly: a part that is there but incomplete earns partial credit, and 0 is kept for a part that is missing. Length: 500 to 900 words of your own writing (interpretation, diagnostics and alt text), not counting code or output. Word counts cover your own writing, not the title, headings, tables, references, figure captions, code, or output, and writing more than the range never costs points. Scores on work that shows effort are adjusted with the course's curve: your final percentage is 25 percent plus 75 percent of your rubric percentage, so 50 of 100 on the rubric becomes 62.5. Work at two-thirds or more of the minimum length counts as showing effort. Attempts are unlimited, so you can revise an early submission right up to the deadline.

- The calls these tables and figures lean on, all of which run the same in VS Code: `count()` for a frequency table, `pivot_wider()` to lay a cross-tabulation out, `fct_reorder()` to order the bars, `labs()` for the axis labels, title, and caption, `scale_fill_brewer()` or the v2v palette for the fill, and position = "fill" with `scales::percent_format()` for a proportion chart. The Describe group of the R Companion's Write R code writes these same calls, including Line chart over time and Histogram comparing two groups, saves each figure with `ggsave()`, lays the crosstab out with `count()` and `pivot_wider()`, leaves an alt-text line marked YOUR TURN in each chart for you to write, and opens the saved chart in a VS Code tab.
- Recommended, not graded here: open the script with the settings block Inferencing Data requires (load the data, rename the variables you use to a fixed set, and make the categorical ones factors), so that next week's script is this one with four tests added.
- What it feeds: your White Paper, which runs these same table and figure calls on your own variables. The Twitch figures stay here; the paper shows only what your own data shows.

Describing Data in R, 100 points

Criterion	Exemplary	Proficient	Developing	Beginning

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Tables and the three numbers 20 points	**20 points.** Frequency tables for the key categorical variables, built with `count()`, and the mean, median, and standard deviation for every continuous variable you report.	**15 to 19 points.** All there, but a table is unlabeled or one variable is missing a number.	**10 to 14 points.** The mean alone, or tables for only some variables.	**0 to 9 points.** None.
Figures 20 points	**20 points.** At least two ggplot2 figures plus one figure of your own choice, labeled, titled, and captioned, saved to a file by the code that makes them, with one sentence on why you chose the third figure for that variable.	**15 to 19 points.** The figures are there, but one is missing a label, the title, or the caption, or the figure you chose has no reason given.	**10 to 14 points.** One figure, or figures saved by hand instead of by the code.	**0 to 9 points.** No figures.
Cross-tabulation 10 points	**10 points.** A cross-tabulation of two categorical variables, laid out with `count()` and `pivot_wider()`, labeled so a reader can tell the rows from the columns. This is the table next week's chi-square tests.	**8 to 9 points.** Present, but the labels are missing.	**5 to 7 points.** Built on the wrong columns.	**0 to 4 points.** Missing.
Alt text 5 points	**5 points.** Every figure has alt text that says what the figure shows.	**4 points.** Alt text on every figure, but one repeats the caption.	**3 points.** Alt text on some figures only.	**0 to 2 points.** None.

Criterion	Exemplary	Proficient	Developing	Beginning
Diagnostics behind each figure 25 points	**25 points.** For each figure: the shape of the variable, whether the assumptions of the tests you are heading toward look reasonable, and where the outliers are, with the check shown.	**19 to 24 points.** Shape and outliers reported, but the assumptions are not tied to the tests you plan to run.	**13 to 18 points.** Diagnostics for some figures only, or stated without being shown.	**0 to 12 points.** Figures without diagnostics. The figure is the claim; the diagnostic is whether you are allowed to make it.
Interpretation 15 points	**15 points.** Two to three sentences after each output.	**11 to 14 points.** Sentences after each output, but they describe the output without saying what it cannot show.	**8 to 10 points.** Interpretation for some outputs only, or a single line each.	**0 to 7 points.** None.
Display choices disclosed 5 points	**5 points.** Every display choice disclosed: bin width, axis caps, merged categories.	**4 points.** One choice is not disclosed.	**3 points.** One disclosure across all the figures.	**0 to 2 points.** None.